

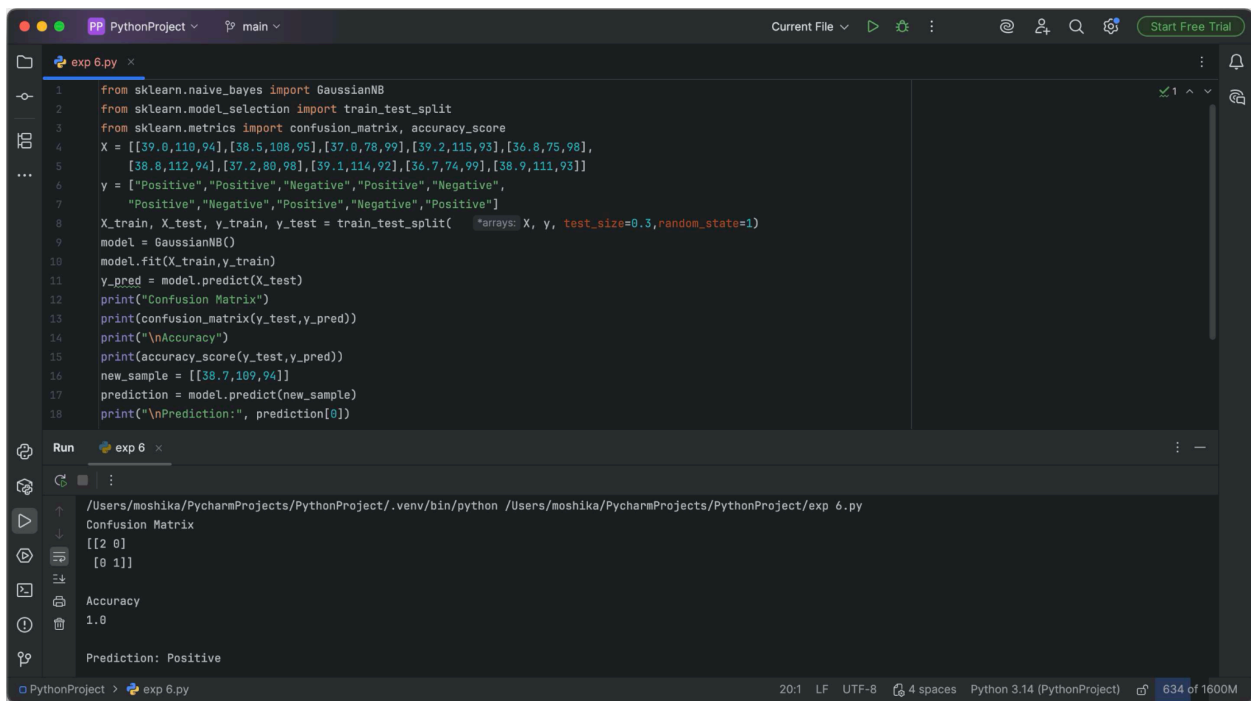
EXPERIMENT - 6

Aim: To implement the Naïve Bayes classification algorithm in Python and evaluate its performance using a confusion matrix and accuracy.

Procedure:

1. Import the required Python libraries.
2. Load the dataset.
3. Separate the input features and target class.
4. Split the dataset into training and testing data.
5. Create a Naïve Bayes classifier.
6. Train the classifier using the training data.
7. Predict the classes for the test data.
8. Generate the confusion matrix.
9. Calculate the classification accuracy.
10. Display the confusion matrix and accuracy.

Output:



```
1 from sklearn.naive_bayes import GaussianNB
2 from sklearn.model_selection import train_test_split
3 from sklearn.metrics import confusion_matrix, accuracy_score
4 X = [[39.0,110,94],[38.5,108,95],[37.0,78,99],[39.2,115,93],[36.8,75,98],
5      [38.0,112,94],[37.2,80,98],[39.1,114,92],[36.7,74,99],[38.9,111,93]]
6 y = ["Positive","Positive","Negative","Positive","Negative",
7      "Positive","Negative","Positive","Negative","Positive"]
8 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)
9 model = GaussianNB()
10 model.fit(X_train, y_train)
11 y_pred = model.predict(X_test)
12 print("Confusion Matrix")
13 print(confusion_matrix(y_test, y_pred))
14 print("\nAccuracy")
15 print(accuracy_score(y_test, y_pred))
16 new_sample = [[38.7,109,94]]
17 prediction = model.predict(new_sample)
18 print("\nPrediction:", prediction[0])
```

Run

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp_6.py
Confusion Matrix
[[2 0]
 [0 1]]
Accuracy
1.0
Prediction: Positive
```

Result: The Naïve Bayes classifier successfully classified the test data. The confusion matrix and accuracy score were displayed to evaluate the performance of the model.